# STACKS, DISCRIMINANTS: LEMMA-NORM-DIFFERENT-IN-DISCRIMINANT

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 1862-1867 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 1869-1904, the complete author proof (34 body lines).

*Editorial note.* Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 20 of queue_discriminant.json (pb=1295, stmt 1862-1867, proof 1869-1904).

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 0BD2: sources/stacks-discriminant/pinned-original-source.tex, source lines 1904–1904. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0BVH: sources/stacks-discriminant/pinned-original-source.tex, source lines 694–759. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $f : Y \to X$ be a finite flat morphism of Noetherian schemes. Then $\mathrm{Norm}_f : f_*\mathcal{O}_Y \to \mathcal{O}_X$ maps $f_*\mathfrak{D}_f$ into the ideal sheaf of the discriminant $D_f$.*

*Proof.* The norm map is constructed in Divisors, Lemma 0BD2 and the discriminant of $f$ in Section 0BVH. The question is affine local, hence we may assume $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$ and $f$ given by a finite locally free ring map $A \to B$. Localizing further we may assume $B$ is finite free as an $A$-module. Choose a basis $b_1, \ldots, b_n \in B$ for $B$ as an $A$-module. Denote $b_1^\vee, \ldots, b_n^\vee$ the dual basis of $\omega_{B/A} = \mathrm{Hom}_A(B, A)$ as an $A$-module. Since the norm of $b$ is the determinant of $b : B \to B$ as an $A$-linear map, we see that $\mathrm{Norm}_{B/A}(b) = \det(b_i^\vee(bb_j))$. The discriminant is the principal closed subscheme of $\mathrm{Spec}(A)$ defined by $\det(\mathrm{Trace}_{B/A}(b_ib_j))$. If $b \in \mathfrak{D}_{B/A}$ then there exist $c_i \in B$ such that $b \cdot b_i^\vee = c_i \cdot \mathrm{Trace}_{B/A}$ where we use a

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dot to indicate the $B$-module structure on $\omega_{B/A}$. Write $c_i = \sum a_{il} b_l$. We have

$$\begin{aligned} \mathrm{Norm}_{B/A}(b) &= \det(b_i^\vee(bb_j)) \\ &= \det((b \cdot b_i^\vee)(b_j)) \\ &= \det((c_i \cdot \mathrm{Trace}_{B/A})(b_j)) \\ &= \det(\mathrm{Trace}_{B/A}(c_i b_j)) \\ &= \det(a_{il}) \det(\mathrm{Trace}_{B/A}(b_l b_j)) \end{aligned}$$

which proves the lemma. $\square$